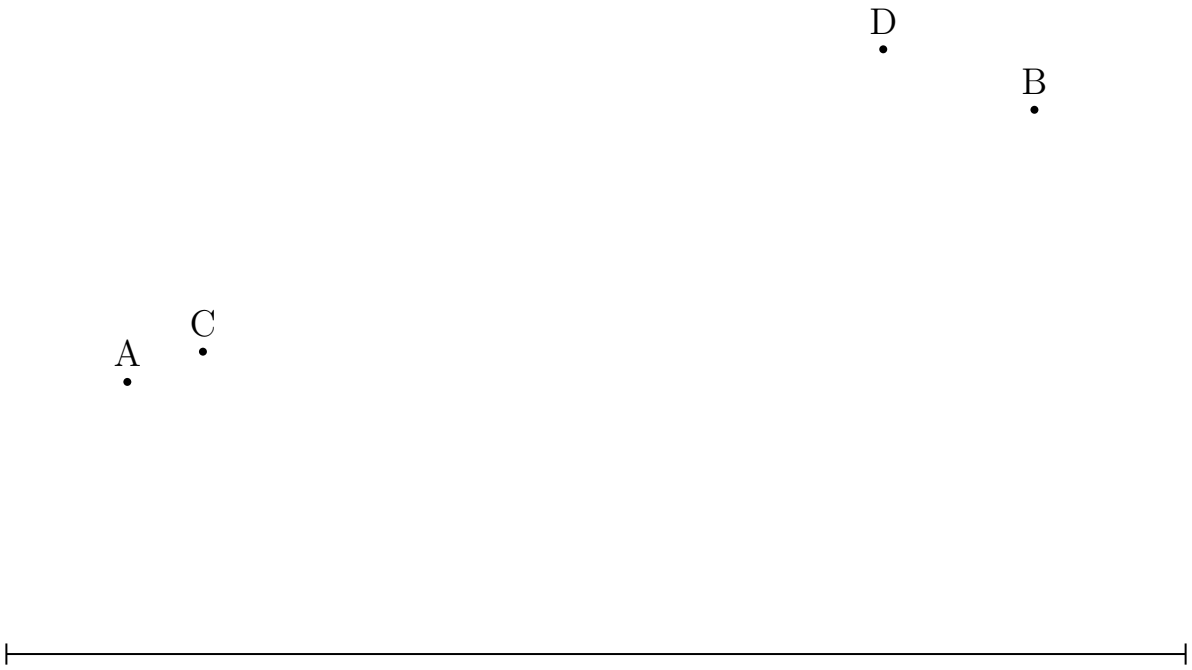
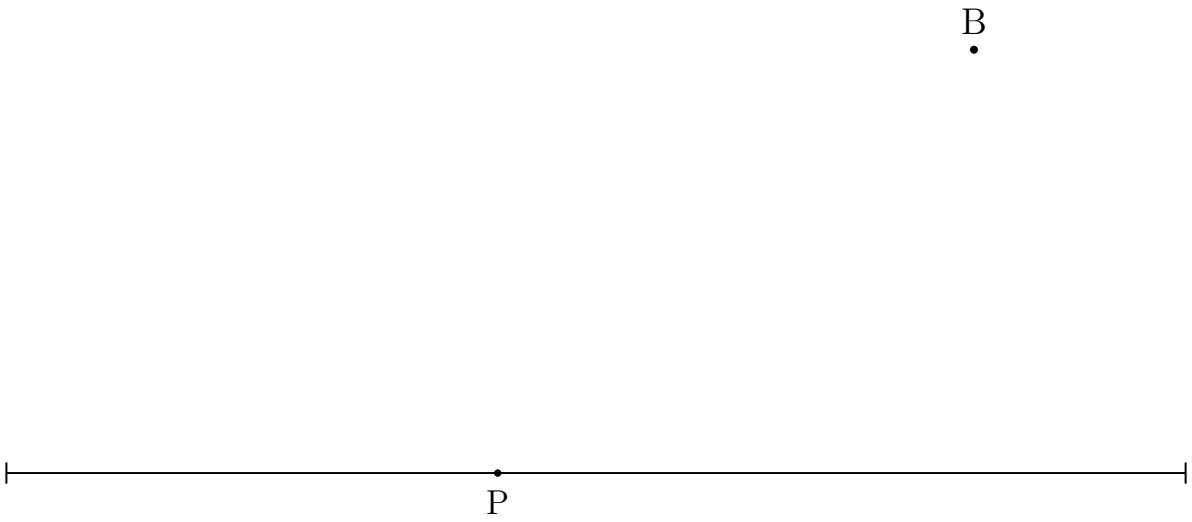
A route goes straight from its start dot to one point on the line, then straight to its finish dot. Touch between the end marks. Do not travel along the line. Use the dot centers.

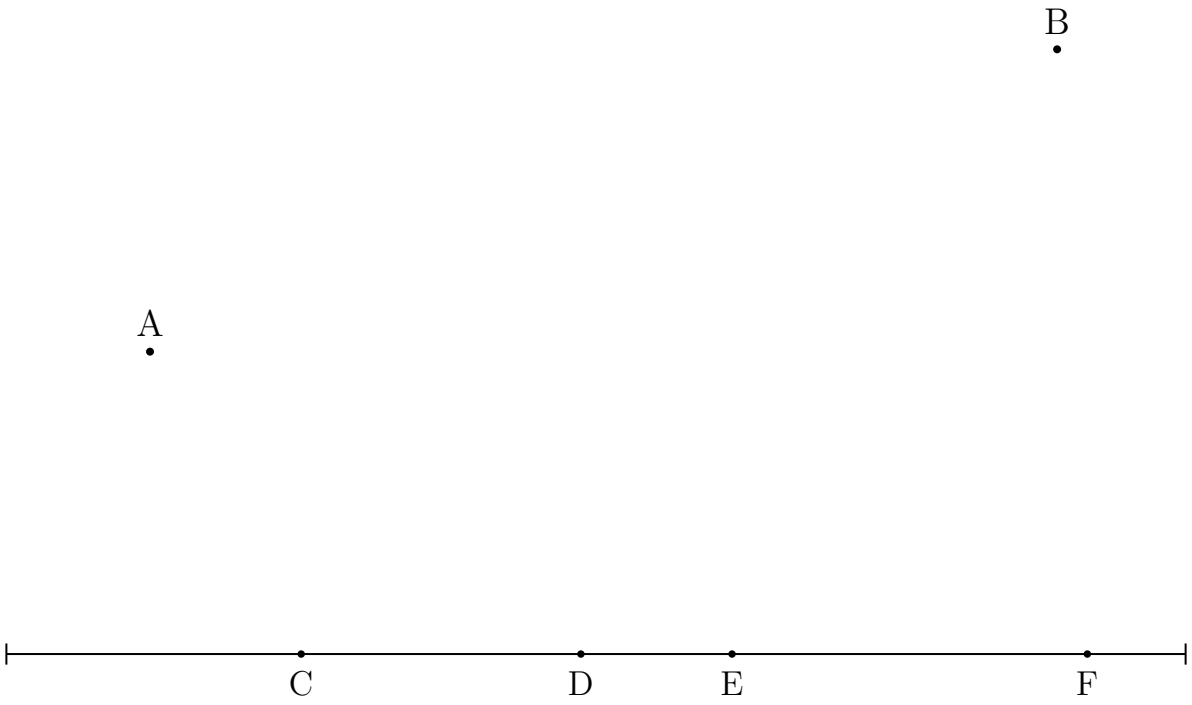
Problem 4: Find the shortest route from A to B and the shortest route from C to D. Can both shortest routes touch at the same point? Explain.



Problem 5: Choose two different start dots above the line so that their shortest routes to B both touch P. Describe all the places above the line where a start dot could go, and explain why.



Problem 6: Find the shortest route from A to B. Now allow touching only between C and D, only between E and F, or only between C and F. In which cases does the same route still give a shortest allowed route? Explain.



Problem 7: Can two different contact points both give the shortest route from A to B? Explain why your answer covers every possible contact point.

